

Nuclear energy between peaceful and military use



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Abstract:

The importance of nuclear energy as a source of energy has increased with the increasing awareness of the benefits of this energy and economic source on all aspects of life, especially in light of the instability of oil prices and the need to search for an alternative source of energy, Researchers are divided about this source between supporters and opponents, because its use is not limited to peaceful purposes, but is also directed to military use, The problem centered on the nature and as for the hypothesis, it suggested that the use of nuclear energy has two main dimensions. The first is civilian uses. as for the second aspect, it is represented in the military field, whose dangers exceeded all expectations. However, the peaceful side is also not without risks. Therefore, the use of nuclear energy entails great responsibilities, especially regarding the responsibility of nuclear safety and security.

Keywords: nuclear energy; Peaceful use; Military use; Nuclear waste; Nuclear security.

المخلص:

ازدادت أهمية الطاقة النووية كمصدر من مصادر الطاقة مع تزايد الوعي بفوائد هذا المصدر الطاقوي والاقتصادي على جميع نواحي الحياة خاصة في ظل عدم استقرار أسعار البترول وضرورة البحث عن مصدر بديل للطاقة، وينقسم الباحثون إزاء هذا المصدر بين مؤيد ومعارض، لأن استخدامه لا يقتصر على الأغراض السلمية بل وكذلك يوجه للاستخدام العسكري، فتمحورت الإشكالية حول طبيعة استخدام الطاقة النووية وأبعادها، أما الفرضية اقترحت أن استخدام الطاقة النووية له بعدين رئيسيين الأول هو الاستعمالات المدنية أما الجانب الثاني فيتمثل في الميدان العسكري والذي فاقت مخاطره كل التوقعات، إلا أن الجانب السلمي لا يخلو كذلك من المخاطر، لذلك فإن استخدام الطاقة النووية يترتب عليه مسؤوليات كبيرة خاصة فيما يتعلق مسؤولية الأمان والسلامة النووية.

الكلمات المفتاحية: الطاقة النووية، الاستخدام السلمي، الاستخدام العسكري، النفايات النووية، الأمن النووي.

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Introduction:

States harness possibilities and means foremost of which is military force To maintain security, This prompted countries to adopt a policy of building strong armies equipped with the latest weapons and military equipment, Especially after the second half of the twentieth century witnessed great scientific progress that was placed at the disposal of mankind with advanced technology, which was not only used positively, Rather, man is able to annihilate the entire world several times because of weapons of mass destruction, whether chemical, biological or nuclearThe destructive capacity of this type appeared, and with it international efforts aimed at limiting its spread.

Nuclear energy is the energy that is released by controlling the reactions of fission or fusion of atomic nuclei, This energy is used in nuclear power plants to heat water to produce water vapor, which is then used to produce electricity, It is a basic and inexhaustible source of energy, but in return for that, its costs are high, in addition to what is left behind by its radioactive waste and safety measures to secure it from risks, Which prompted thinking about systems of safeguards for the peaceful use of nuclear energy.

- **What are the different uses of nuclear energy and what are their implications?**

This problem leads us to a set of sub-questions:

What is nuclear energy and how was it discovered?

What are the dimensions of nuclear energy?

To answer the main problem and its sub-questions, we try to test the following central **hypothesis**:

Nuclear energy is used for civilian purposes as it is directed towards military uses, and both uses result in material and immaterial threats.

Considering method is the basic rule for all scientific research, in view of the nature of the subject each of the analytical descriptive approaches were adopted as well as some steps of the historical approach, in the element related to the emergence and development of nuclear energy.

This research paper includes **three main axes**:

- The discovery and concept of nuclear energy.
- Different uses of nuclear energy and its effects.
- Advantages and Disadvantages of nuclear power in peacetime.

1. The discovery and concept of nuclear energy:

There are two main categories of energy sources, which are the traditional non-renewable sources and non-traditional or renewable sources such as solar energy, wind energy, wave energy and nuclear energy.

Nuclear energy is a form of energy released from the nucleus, the core of atoms, made up of protons and neutrons. This source of energy can be produced in two ways: fission – when nuclei of atoms split into several parts – or fusion – when nuclei fuse together (Galindo, 2022).

a. The discovery of nuclear energy:

The Greek scientist **Democritus**(460AC-370AC) is the first to give an alternative concept of matter, and that matter of all kinds consists of bodies that can be divided until they reach single particles that cannot be divided, they are so small that particles smaller than them cannot be imagined Each of these particles has been called **an atomos** In Greek, it means the inseparable part.

At the end of the 19th century, a majority of scientists thought that there was not much left to the history of nuclear energy began by chance on March **1896**. discover in the field of physics. ” The atom did not exist”: some academics even said so (Jean-Claude, 2021) .

So, the history of nuclear energy began by chance on March **1896**, when **Henri Becquerel** discovered that the blackening of the photographic plate occurred even when the salt was not exposed to sunlight, **Marie Curie** and her husband **Pierre** studied the phenomenon and isolated two new elements that exhibited this spontaneous energy production: Polonium and Radium. They named the phenomenon radioactivity.

In England, **Ernest Rutherford** starts studying radioactivity and discovers that there are two types of rays that come out that are different from x-rays. He calls them alpha- and beta-radiation. He later discovers the shocking and appalling fact that the vast majority of the mass of atoms is concentrated in their centers, and thus discovers the atomic nucleus. He is widely regarded today as the father of nuclear physics. He later discovers gamma radiation. In **1920**, he theorizes the existence of a neutral particle in the nucleus called a neutron, though there is no evidence that neutrons exist yet, in **1932**, **Chadwick** reads some published results from the Curie’s kid, Irene Joliot-Curie that says gamma radiation was found to knock protons out of wax. Disbelieving, he suspects they are seeing Rutherford’s neutrons and does experiments to prove this, thus discovering the neutron (Touran, 2013).

Breit and **Wigner** provided the theoretical explanation of slow neutron processes in **1936**. **Fermi** made measurements of the distribution of both fast and thermal neutrons and explained the behavior in terms of elastic scattering, chemical binding effects, and thermal motion in the target molecules. During this period, many cross sections for neutron reactions

were measured, including that of uranium, but the fission process was not identified, **Fermi** then suggested that neutrons might be emitted during the process, and the idea was born that a chain reaction that releases great amounts of energy might be possible. As early as July 1939, four leading scientists-Szilard, Wigner, Sachs, and Einstein-had initiated a contact with President Roosevelt, explaining the possibility of an atomic bomb based on uranium. (Raymond L. Murray, 2000)

An experiment at the University of Chicago was crucial to the success of the Manhattan Project and also set the stage for future nuclear developments. On December 2, **1942**, the system was ready. At Los Alamos, New Mexico a research laboratory was established under the direction of J. Robert Oppenheimer. Theory and experiment led to the development of the nuclear weapons, first tested at Alamogordo, New Mexico, on July 16, 1945, and later used at Hiroshima and Nagasaki in Japan (Raymond L. Murray, 2000).

b. Definition of Nuclear energy:

Energy is defined as the ability to do something, and it exists in several forms, all of which are related to movement, and energy can move from one body to another and from one form to another, as it is classified depending on the new form to which it moved, and it cannot be generated, or destroyed, and this is what is known as the law of conservation of energy, or the first law of thermodynamics.

All of the activities of human beings depend on energy, as we realize when we consider the dimensions of the world's energy problem. The efficient production of food requires machines, fertilizer, and water, each using energy in a different way. Energy can exist in many forms as Thermal energy, radiant energy, chemical energy, electrical energy, kinetic energy, acoustic energy, elastic energy, gravitational energy, and nuclear energy. It is the subject of this research paper.

Nuclear energy was defined as the energy resulting from changes in atomic nuclei, or nuclear reactions, and the development of nuclear energy provided a new source of energy, as it is possible to take advantage of the heat of the nuclear reactor to produce steam that is directed to operate electric generators, or propellers of large ships, or other machines by using a turbine, and the United States production of electricity from nuclear reactors constituted **23%** in **1999**.

Nuclear energy is the energy released by a chain of reaction, specifically by nuclear fission or fusion in the reactor

Below we will present some definitions about nuclear energy:

- The energy released by a nuclear reaction (either fission or fusion) or by radioactive decay. (Davies, 2018)

- nuclear energy, also called atomic energy, energy that is released in significant amounts in processes that affect atomic nuclei, the dense cores of atoms, It is distinct from the energy of other atomic phenomena such as ordinary chemical reactions, which involve only the orbital electrons of atoms. (encyclopedia, 2023)
- In nuclear fission, atoms are split apart to form smaller atoms, releasing energy. Fission takes place inside the reactor of a nuclear power plant. At the center of the reactor is the core, which contains uranium fuel. (Information, 2022)

2. Different uses of nuclear energy and its effects:

a. The peaceful dimension of nuclear energy:

The **1968** Non-Proliferation Treaty granted all States parties the inalienable right to produce and use nuclear energy for peaceful purposes, and that right was set out in the preamble as well as in some articles of the Convention, which considered the peaceful use of nuclear energy as a peaceful activity carried out by countries and does not aim to possess or manufacture nuclear weapons or possess explosive devices.

It should be noted that the Nuclear Non-Proliferation Treaty not specified Meaning of the peaceful uses of nuclear energy, and The International Atomic Energy Agency (IAEA) is a key facilitator in the development and delivery of the peaceful uses of nuclear energy and technology. The United States is by far the largest contributor to IAEA peaceful use programs, providing more than \$395 million since 2015. (Nonproliferation, 2022)

Security Council Resolution **No. 1696** of **July 31, 2006** included the need for countries to adhere to the Treaty on Research, Production and Use of Nuclear Energy for Peaceful Purposes, Such as the development of scientific research and electricity production centers and radioactive isotopes in the areas of development, And its use in the fields of agriculture, transportation, urbanization and hydrology.

Firstly: the use of radioactive isotopes in scientific research and development

Radionuclide is an unstable form of a chemical element that releases radiation as it breaks down and becomes more stable. Radionuclides may occur in nature or be made in a laboratory. In medicine, they are used in imaging tests and in treatment. Also called radioisotope (Institute)

So, we use nuclear energy in Radiation Therapy which is X-ray treatment that damages or kills cancer cells (Institute, The NCI Consumers' Cancer Dictionary for Peer Review, 2009)

For six decades, **Berkeley** Laboratory has been acknowledged as the birthplace of nuclear medicine, Budinger, who heads the Center for Functional Imaging, chronicled the preeminent role of the Lab in nuclear medicine -- in the diagnosis and treatment of diseases, in imaging, and in safety, they have provided physicians new and more effective ways to treat diseases. And they have devised treatments for diseases that had been untreatable. (Khan, 1996)

Space Exploration:

A great deal of what we know about deep space has been made possible by **radioisotope power systems (RPSs)**. These small nuclear power sources are used to power spaceships in the extreme environments of deep space, RPSs are proven to be safe, reliable, and maintenance-free for decades of space exploration, including missions to study Jupiter, Saturn, Mars, and Pluto. (Energy, 2018)

In the field of space, nuclear fuel is used to operate spacecraft and to launch missiles and satellites.

Criminal Investigation:

Criminal investigators frequently rely on radioisotopes to obtain physical evidence linking a suspect to a specific crime. They can be used to identify trace chemicals in materials such as paint, glass, tape, gunpowder, etc. In the field of agriculture, environment and hydrology and poisons. (Ibid)

Secondly: In the field of agriculture, environment and hydrology

Nuclear technology and biodiversity are an added value to plant and animal genetic resources, The presence of abundant genetic resources in any ecoregion is essential for sustainable agriculture and food production.

Applications of isotopes and radiation in agriculture have been one of the most important areas of technical co-operation at the IAEA. During the last decade, about 20% of all disbursement related to this field, making it the leading area of activity in the IAEA's technical co-operation programme, In the Asia and Pacific region, the field of agriculture is one of the three most important areas of activity, following those related to applications of isotopes in industry and hydrology and to nuclear engineering and technology. The Agency is currently executing 48 projects in 13 of the 15 Member States of the region which receive IAEA technical assistance. (L. LaChance, 1990)

B. Military use of nuclear energy:

Uranium enrichment is the sensitive point that separates the peaceful and military use of nuclear, Noting that the statute of the International Atomic Energy Agency did not precisely define the meaning of "military purposes", however, the use of nuclear energy in operating naval battleships or lighting military installations with electricity generated by nuclear energy can be considered a non-peaceful use of nuclear energy in addition to the use of that energy in the manufacture of weapons mass destruction.

To understand the military and civilian uses of nuclear technology, it is necessary to explain nuclear fission. Fission means the splitting of an atom. When a fissile isotope absorbs a neutron, it splits into two and yields at least two neutrons. This makes it possible to have a fission chain reaction and releases an enormous amount of energy. The first application of this technology

was the nuclear weapon, which was developed by the United States during World War II and was used against Japan in 1945, ending the war (Udum)

3. Advantages and Disadvantages of nuclear power in peacetime:

The first successful demonstration of the use of nuclear power as a source of electricity production took place in the early 1950s. Since then, the technology of nuclear power reactors and the construction and operation of nuclear power plants have passed through different stages leading to the maturity of several nuclear power reactor systems now in use for large-scale electricity generation. (Agency, 1982)

In the following, we will briefly present the most important advantages of nuclear energy, because they have been detailed in the context of its peaceful uses:

- _ Nuclear power plants don't require a lot of space.
- _ It doesn't pollute (it does, but in a very different way... more about it further on.)
- _ Nuclear energy is by far the most concentrated form of energy.

However, this does not preclude the existence of disadvantages to the use of nuclear energy, even for peaceful purposes, especially when it comes to nuclear accidents in addition to nuclear waste.

Firstly: The most prominent of these nuclear accidents:

1- Three Mile Accident:

In 1979, at the **Three Mile Island** near Harrisburg, Pennsylvania, the cooling system of a nuclear reactor failed. Radiation escaped, forcing tens of thousands of people to run away. Fortunately, the problem was solved minutes before a total meltdown would have occurred, and there was no death (Celina Treviño, Advantages & Disadvantages of Nuclear Energy, 2008) Although the radiation leakage outside the facility, the facility itself is no longer usable and must be closed permanently.

2- Chernobyl reactor disaster:

In 1986, a much worse disaster hit Russia's Chernobyl nuclear power plant. In this incident, a large amount of radiation escaped from the reactor, Hundreds of thousands of people were exposed to the radiation. Several dozen died within a few days. In the upcoming years, thousands more may die of cancers induced by the radiation. (Ibid)

3- Fukushima reactor disaster :

On March 11, 2011, a strong earthquake occurred in eastern Japan and caused tsunamis, which led to a disruption of the electrical network. This sudden and emergency defect prevented the minimum level of cooling within the reactor units, immediately after the Fukushima accident in 2011, radiation levels increased in food, water, and the ocean near the

Fukushima Daiichi plant. Because of the threat of radiation exposure, some 150,000 people were forced to evacuate their homes. There were subsequently also multiple leaks at the facility.

The explosion, along with a fire touched off by rising temperatures in spent fuel rods stored in reactor 4, led to the release of higher levels of radiation from the plant, then a large tsunami washed over the reactor site, knocking out the backup generators. While some batteries remained operable, the entire site lost the ability to maintain normal reactor cooling and water circulation functions (Agency, Fukushima Nuclear Accident Update Log, 2017)

Secondly: Nuclear Waste:

One of the most important topics raised in the field of environmental pollution is waste and nuclear waste because it contains remnants of the use of the atom, whether for peaceful purposes or for military purposes.

Radioactive waste is a broader term that includes all waste that contains radioactivity. Nuclear waste is produced as a result of the activities needed to produce nuclear fission. These include extraction of Uranium from ore, concentration of Uranium, processing into nuclear fuel, and disposal of by-products. (Gareth Davies, op.cit)

Conclusion:

The concept of the atom has existed for many centuries, but we only recently began to understand the enormous power contained in the tiny mass which is known as nuclear energy.

Nuclear energy is the second largest source of low-carbon energy in the world, according to the World Nuclear Energy Association, and it can be used in several fields, most notably electricity generation, given that its scope is broader and its uses more comprehensive than limiting it to destructive weapons.

Also, the use of nuclear energy is not limited to generating electricity. Rather, this energy and its technologies can be used in several fields, such as agriculture, food, medicine, space exploration, and water desalination, nuclear techniques provide images of the inside of the human body, and contribute to the treatment of some diseases, as it enables doctors, according to nuclear research, to determine the amount of radiation needed to kill cancer cells without harming healthy cells.

However, this positive aspect of this energy is not without risks, even if it is used for peaceful purposes, in addition to the need for nuclear energy for capital costs, which include preparing the appropriate site for the establishment of the reactor, and engineering, manufacturing, construction, assignment and financing. It is also a renewable energy that is environmentally friendly, but the raw materials on which it depends, such as uranium, are non-renewable.

Although the amount of environmentally harmful gases produced by nuclear power plants is negligible compared to fossil fuels, they still carry many risks related to human health, and the environment is summarized in nuclear radiation in the event of a leak, as well as human waste that poses a threat to the environment, and living organisms in general.

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